AI Playbook and Toolkit for Public Health Departments

Electronic Case Reporting Technical Workflow

ARC 230

Electronic case reporting is one of the clearest examples of public health automation. It uses data from electronic health records and reporting logic to support more timely case reporting to public health agencies. This module teaches learners how eCR works technically and how AI might support eCR-related workflows without replacing public health authority, epidemiologic review, or system documentation.

Level	applied/foundational
Primary track	Technical Architecture Track
Appears in tracks	technical-architecture

Learning Objectives

- Describe the basic eCR workflow from EHR trigger to public health action.
- Explain the roles of eICR, Reportability Response, eRSD, and RCTC.
- Identify eCR data quality, routing, and onboarding issues that can affect AI use.
- Assess where AI may support eCR workflows and where human review remains required.
- Produce an AI-readiness review for an eCR workflow.

Module Overview

Electronic case reporting is one of the clearest examples of public health automation. It uses data from electronic health records and reporting logic to support more timely case reporting to public health agencies. This module teaches learners how eCR works technically and how AI might support eCR-related workflows without replacing public health authority, epidemiologic review, or system documentation.

Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

eCR matters for AI because it brings structured, timely, clinically relevant information into public health workflows. It can reduce manual reporting burden and improve the speed of public health awareness. However, eCR is not simply a data feed. It includes trigger logic, routing, standards, jurisdictional rules, provider workflows, public health system intake, and case investigation processes.

AI may support eCR by summarizing incoming reports, identifying missing information, prioritizing work queues, flagging possible duplicates, or assisting with routing. These uses can be valuable, but they depend on accurate trigger criteria, complete eICR content, correct reportability logic, and careful human review. An AI tool should not silently decide that a case is irrelevant, close a record, or override public health reporting rules.

Understanding eCR also helps staff distinguish automation from judgment. eCR can automate exchange and reporting logic, but public health professionals still interpret context, apply jurisdictional procedures, communicate with providers, and document final decisions. AI should be designed to support those responsibilities rather than obscure them.

Core Concepts

Electronic case reporting. Electronic case reporting is the automated exchange of case report information from healthcare to public health. It is designed to improve timeliness, completeness, and consistency in reportable condition workflows.

eICR. The electronic initial case report is the structured report sent from healthcare to public health. It may include patient demographics, encounter details, diagnoses, lab results, medications, pregnancy status, provider information, and other clinically relevant data.

Reportability Response. A Reportability Response communicates information back to the reporting system or provider, such as whether a case is reportable and what next steps or guidance may apply. It supports a more interactive reporting process.

eRSD and RCTC. Electronic Reporting and Surveillance Distribution supports the distribution of reporting criteria and trigger code value sets. Reportable Conditions Trigger Codes help EHRs determine when an eICR should be generated. These components are essential to reliable and current triggering.

Public Health Example

A jurisdiction may receive eICRs for a reportable condition and want AI to create an investigator-facing summary. The tool could highlight the suspected condition, key dates, lab evidence, symptoms, provider details, pregnancy status, missing fields, and previous related records. This could reduce staff burden, but only if users can inspect the source data and correct errors.

Another jurisdiction may want to use AI to triage incoming eICRs by urgency. This use has higher risk because prioritization may affect response timing. The agency would need to validate the triage logic, assess false positives and false negatives, review subgroup performance, and ensure that staff can override the AI recommendation.

Technical Deep Dive

The eCR workflow begins before a report reaches public health. EHR systems use trigger criteria and value sets to identify encounters that may need reporting. The eICR is generated and transmitted through an exchange process to the appropriate public health jurisdiction. Public health systems then receive, parse, route, review, and incorporate the information into surveillance workflows.

AI readiness depends on several technical questions. Staff need to know which eICR sections are available, how data are mapped into the surveillance system, whether narrative text is included, how person and case matching occur, and how missing or conflicting information is displayed. They also need to know whether the AI tool uses only the eICR, the surveillance record, prior case history, or external guidance documents.

Validation should include realistic eCR scenarios. Test cases should include complete reports, incomplete reports, duplicate reports, conflicting lab and diagnosis information, reports from different provider types, and cases requiring escalation. The goal is to test whether the AI-supported workflow helps staff review reports more effectively without hiding uncertainty or creating unsafe automation.

Risks, Failure Modes, and Guardrails

Incorrect trigger assumptions. If trigger value sets or reporting criteria are outdated, the workflow may generate missing or inappropriate reports. Guardrails include current eRSD/RCTC management, onboarding checks, and periodic review.

Unsupported summaries. Generative AI summaries may include statements not supported by the eICR. Guardrails include source citations, side-by-side review, and user correction workflows.

Unsafe triage. AI prioritization may delay review of important cases if false negatives occur. Guardrails include validation, monitoring, threshold review, and human override.

Routing errors. Incorrect jurisdiction or program routing can delay response. Guardrails include routing validation, exception queues, and escalation procedures.

Application to AI-Supported Workflows

Generative AI may summarize eICR content, but summaries should display source evidence and uncertainty. Predictive models may support prioritization, but only after validation against public health outcomes and review workflows. RAG may provide disease-specific guidance to investigators, but document sources must be current and authoritative. Agentic AI could create follow-up tasks, but task creation should be constrained and auditable.

Reflection Questions

Where does your agency receive, review, or process case reports today?

Which steps are automated and which remain manual?

Where are timeliness, completeness, routing, or duplication problems most common?

Which eCR tasks could AI support safely?

Which tasks require human review, legal authority, or epidemiologic judgment?

Practical Exercise

Create an AI-readiness review for one eCR workflow. Map the current workflow, identify data elements required for review, describe common data quality gaps, define the AI-supported task under consideration, and specify human review, validation, monitoring, and escalation requirements.

Expected Artifact or Evidence

eCR workflow map

AI-readiness checklist

Validation scenarios

Human review and escalation plan

Expected Assignment

- eCR workflow map
- AI-readiness checklist
- Validation scenarios
- Human review and escalation plan

Knowledge Check

- 1. What is eCR?
- 2. What is an eICR?
- 3. What do eRSD and RCTC support?
- 4. Which AI-supported eCR use requires particularly careful validation?
- 5. What should be documented before using AI in an eCR workflow?

References and Resources

- CDC Electronic Case Reporting resources: https://www.cdc.gov/ecr/php/index.html
- AIMS eCR triggering documentation: https://ecr.aimsplatform.org/ehr-implementers/triggering/
- HL7 eCR FHIR Implementation Guide: https://hl7.org/fhir/us/ecr/
- Jurisdictional eCR onboarding and surveillance documentation